Soren L. Larsen
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Curriculum Vitae

ACADEMIC APPOINTMENTS

Postdoctoral Scholar, Department of Demography, University of California, Berkeley, May 2025 - present
Advised by Dr. Ayesha S. Mahmud.

EDUCATION

Doctor of Philosophy, Ecology, Evolution, and Conservation Biology, University of Illinois at Urbana Champaign, IL, Jan. 2021 - May 2025
Advised by Dr. Pamela P. Martinez (School of Molecular and Cellular Biology, Department of Statistics). Dissertation title: Host heterogeneity and respiratory disease dynamics: from structural inequity to immune history.
Master of Science, Mathematics, University of Illinois at Urbana Champaign, IL, Jan. 2021 - May 2022
Advised by Dr. Pamela P. Martinez.
Bachelor of Science, Mathematics, University of Washington, Seattle, WA, Sep. 2016 - Jun. 2020
Magna Cum Laude. Minors in Norwegian and the Comparative History of Ideas.

ADDITIONAL RESEARCH EXPERIENCE

Research Collaboration, COVID-19 Scenario Modeling Hub, May 2024 - present

- *Round 19:* Calibrated model to historical COVID-19 data and contributed projections for 2025-26. Co-wrote manuscript and conducted ensemble analysis with other modeling team leads.
- *Heterogeneity research round*: Mentored by Cecile Viboud (Fogarty International Center, NIH). Calibrated model and contributed projections for a Scenario Modeling Hub research round led by Samantha Bents (Stanford) on demographic heterogeneity in COVID-19 transmission and severity.

MARTIANS Student Intern, Sandia National Labs, May 2023 - Aug. 2023

- Worked with the epidemiology group led by Patrick Finley, exploring outcomes from socioeconomically-explicit vs. -agnostic models of SARS-CoV-2 transmission.

Complexity-GAINS International Summer School (Santa Fe Institute), Vienna, Austria, July 2022

- Two week, NSF-funded workshop on disintegration of societies: quantitative modeling of complex socio-behavioral systems. Co-organized by the Santa-Fe Institute.

UNDERGRADUATE MENTEES

Rita Liao, UC Berkeley, 2026 - present
Ria Cherukuri, Integrative Biology, UIUC, 2026 - present || **Julie Li,** Statistics, UIUC, 2026-present
Paige Masi, Ecology and Evolutionary Biology, Princeton, 2026 - present
Alyssa Anastasi, Statistics, UIUC, 2025 - present || **Chloe Yang**, Statistics, UIUC, 2023 - present
Lydia Schieleit, Statistics, UIUC, 2024 || **Annika Venegas**, Mathematics, UIUC, 2023 - 2024
Erin Haslett, Political Science, UIUC, 2023 - 2024
Haylee West, Molecular and Cellular Biology, UIUC, 2021 - 2023
Jefrin Joseph, Molecular and Cellular Biology, UIUC, 2021 - 2022 || **Ikgyu Shin**, Statistics, UIUC, 2022
Eliana Chandra, Statistics/Sociology, UIUC, 2022 || **Rafael Anorga**, Integrative Biology, UIUC, 2022

TEACHING EXPERIENCE

Guest Lecture, STAT 430, University of Illinois at Urbana-Champaign, March 5, 2026

- Lecture on serocatalytic modeling methods for infectious diseases, with an interactive lab in R.

Teaching Assistant, University of Illinois at Urbana-Champaign, Jan. 2021 - Dec. 2022

- Fall 2022: one Merit discussion section of Calculus 2 (Math 231). Wrote worksheets/mock exams and facilitated complex problem solving and collaboration.
- Spring 2022: two discussion sections of Calculus 2 (Math 231). Graded worksheets and helped develop exams.
- Fall 2021: two discussion sections of Calculus I (Math 221). Graded worksheets and exams.
- Spring 2021: two discussion sections of Calculus for Business (Math 234)

Math Grader, University of Washington, Winter/Summer Quarters 2019, Summer Quarter 2020

- Summer 2020: Graded homework for one section of Linear Analysis (MATH 309)
- Summer 2019/2020: Graded homework with feedback for one section of Introduction to Mathematical Reasoning (MATH 300) each summer
- Winter 2019: Wrote homework keys and graded homework for two sections of Linear Analysis (MATH 309)

Volunteer Peer-Mentor, University of Washington, Spring/Fall 2018, Fall 2019

- Peer-mentors are part of the UW STEM-Dawgs program, whose purpose is to foster a sense of community within large undergraduate lecture courses, and address disparities in performance for women, underrepresented groups, and first-generation college students.
- Spring 2018: Trained in research-based pedagogical methods
- Fall 2018: Facilitated problem-solving and study skill development in a weekly discussion workshop for General Chemistry (CHEM 142), under the supervision of a graduate teaching assistant.
- Fall 2019: Facilitated an additional workshop with a graduate student

PEER-REVIEWED PUBLICATIONS (***** indicates corresponding)

- Bents, S.J., **Larsen, S.L.,** Jung, S., Howerton, E., Contamin, L., Loo, S.L., Gozzi, N., Davis, J.T., Vespignani, A., Chinazzi, M., Perra, N., Bandekar, S.R., Bi, K., Bouchnita, A., Fox, S.J., Meyers, L.A., Srivastava, A., Al Aawar, M., Chen, J., Venkatramanan, S., Hoops, S., Lewis, B., Bhattacharya, P., Machi, D., Marathe, M., Pei, S., Yao, Q., Yang, Q., Hill, A., Lemaitre, J., Nande, A., Willard, T., Pearson, C.A.B., Przykucki, E., McKee, C., Macdonald, J., Alleman, T., Smith, C.P., Carcelen, E., Yan, K., Hochheiser, H., Viboud, C., Lessler, J., Shea, K., León, T.M., Truelove, S. (in press). Multi-model insights into racial and ethnic heterogeneities in early COVID-19 transmission and severity in the U.S. *American Journal of Epidemiology*.
- Liu, H., Yang, J., **Larsen, S. L.**, Martinez, P. P., & Dayanikli, G. (2026). Incorporating Authority Perception, Economic Status, and Behavioral Response in Infectious Disease Control. *Journal of the Royal Society Interface, 23*(244), 20260281.
- **Larsen, S. L.***, Yang, J., Lv, H., Huan, Y. W., Teo, Q., Pholcharee, T., Lei, R., Gopal, A.B., Shao, E.K., Talmage, L., Mok, C.K.P, Takahashi, S., Kraay, A.N.M., Wu, N.C., & Martinez, P. P.* (2025). Reimagining the serocatalytic model for infectious diseases: a case study of common coronaviruses. *Epidemics, 53,* 100859.
- **Larsen, S.L.,** Noor, I., West, H., Chandra, E., Martinez, P. P., Kraay, A. N. M. (2025). Influence of immune history when choosing a SARS-CoV-2 booster strategy. *Scientific Reports, 15*(1), 35640.
- Kostandova, N., Schluth, C., Arambepola, R., Atuhaire, F., Bérubé, S., Chin, T., Cleary, E., Cortes-Azuero, O., Garcia-Carreras, B.,Grantz, K.H., Hitchings, M.D.T., Huang, A.T., Kishore, N., Lai, S., **Larsen, S.L.**, Loisate, S., Martinez, P., Meredith, H.R., Purbey, R., Ramiadantsoa, T., Read, J., Rice, B.L., Rosman, L., Ruktanonchai, N., Salje, H., Schaber, K.L., Tatem, A.J., Wang, J., Cummings, D.A.T., Wesolowski, A. (2024). A systematic review of using population-level human mobility data to understand SARS-CoV-2 transmission. *Nature Communications*, *15*(1), 1-12.
- **Larsen, S. L.**, & Kraay, A. N. (2024). Transparent transmission models for informing public health policy: the role of trust and generalizability. *Proceedings of the Royal Society B*, 291(2015), 20232273.
- **Larsen, S. L.**, Shin, I., Joseph, J., West, H., Anorga, R., Mena, G. E., Mahmud, A.S., & Martinez, P. P. (2023). Quantifying the impact of SARS-CoV-2 temporal vaccination trends and disparities on disease control. *Science Advances*, *9*(31), eadh9920.

PREPRINT/UNDER REVIEW (***** indicates corresponding, ^ indicates shared first author)

- **Larsen, S. L.*,^**, Yang, J.^, Haslett, E. M., Anastasi, A., Venegas, A., Schieleit, L., Mahmud, Ayesha S., & Martinez, P. P.* (2026). Same household, different choices: variation in health behaviors related to respiratory viruses in Illinois. *medRxiv*, 2026-05.
- **Larsen, S. L.***, Martinez, P. P., & Mahmud, A. (2026). Changing COVID-19 vaccine eligibility could reshape disease burden for all. *medRxiv*, 2026-04.
- Nande, A., **Larsen, S.L.**, Turtle, J., Davis, J.T., Ramdas Bandekar, S., Lewis, B., Chen, S., Contamin, L., Jung, S., Howerton, E., Shea, K., Bay, C., Ben-Nun, M., Bi, K., Bouchnita, A., Chen, J., Chinazzi, M., Fox, S.J., Hill, A.L., Hochheiser, H., Lemaitre, J.C., Loo, S.L., Marathe, M., Meyers, L.A., Pearson, C.A.B., Porebski, P., Przykucki, E., Smith, C.P., Venkatramanan, S., Vespignani, A., Willard, T.C., Yan, K., Viboud, C., Lessler, J., Truelove, S. (2026) Assessing the Impact of Timing and Coverage of United States COVID-19 Vaccination Campaigns: A Multi-Model Approach. *medRxiv preprint*

2026.04.07.26349269.

OTHER PUBLICATIONS

- **Larsen, S.L.,** Noor, I., West, H., Chandra, E., Martinez, P.P., Kraay, A.N.M. "Disentangling the impact of natural infections and vaccination in a rapidly evolving pandemic: prioritizing limited vaccine supplies in low- and middle-income countries" WHO pink book. Meeting of the Advisory Committee on Immunization and Vaccines related Implementation Research. February 2023, pages 215-271.
- **Larsen, S. L.**, Beyeler, W. E., Acquesta, E. C. S., Klise, K. A., & Finley, P. D. (2023). *Socioeconomically-inspired modeling to justify use of fine-grain mobility data* (No. SAND2023-07730). Sandia National Lab.(SNL-NM), Albuquerque, NM (United States).

GRANTS

SHIELD Illinois, 2023-2025 (Co-Investigator), $100,000
Disentangling the effect of racial, ethnic, and socioeconomic health disparities on respiratory disease transmission in Illinois before and during the COVID-19 pandemic

HONORS AND AWARDS

Isabel Norton Award, Spring 2024
PEEC Travel Grant, Spring 2024
Best Talk, Annual Microbiology Research Conference, Monticello, Illinois, October 2023
List of Teachers Ranked as Excellent, Spring 2021 - Fall 2022
Barbara Sando Scholarship in Mathematics, awarded for Winter/Spring 2020
Phi Beta Kappa, inducted Spring 2019
Annual Dean's List, University of Washington, 2016-17 and 2017-18

TALKS AT MEETINGS, CONFERENCES AND SEMINARS

Mathematical and Statistical Modeling of Spread and Mitigation of Epidemics, Institute of Mathematical and Statistical Innovation, University of Chicago, May 2026
Invited participant - 25 minute research talk.
Inequalities in Infectious Disease Dynamics Symposium, London School of Hygiene and Tropical Medicine, April 2026
Sociodemographic disparities in health seeking behavior for respiratory viruses - 7 minute research talk at a public symposium. Additionally participated in a closed two-day workshop on social inequalities and infectious disease modeling.
Berkeley Population Biology of Infectious Diseases Seminar Series, February 2026
Understanding the role of sociodemographic disparities in shaping respiratory disease dynamics - 25 minute research talk
Berkeley Demography Brownbag Series, February 2026
Understanding the role of social disparities in shaping respiratory disease dynamics - 40 minute research talk
Scenario Modeling Hub Annual Meeting, September 2025
Presentation of Round 19 results with co-authors - 15 minute research talk
Epidemiology and Population Studies modeling seminar, NIH Fogarty International Center, Bethesda, Maryland, February 2025
Reimagining the serocatalytic model for infectious diseases: a case study of common coronaviruses - 50-minute research talk
Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting, Silver Springs, Maryland, November 2024
Reimagining the serocatalytic model for seasonal coronaviruses - 5 minute speed talk
Brook Lab, University of Chicago, Chicago, Illinois, November 2024
Host heterogeneity and SARS-CoV-2 dynamics: from structural inequity to immune history - 50-minute research talk
Epidemiology and Population Studies modeling seminar, NIH Fogarty International Center, Bethesda, Maryland, October 2024
Host heterogeneity and SARS-CoV-2 dynamics: from structural inequity to immune history - 50 minute research talk
Bohrer Memorial Student Workshop in Statistics, April 2024
Immune history influences SARS-CoV-2 booster impacts - 15-minute research talk and poster presentation
Microbe Mondays, April 2024
Host heterogeneity and SARS-CoV-2 dynamics: from structural inequity to immune history - 20-minute research talk

Annual Microbiology Research Conference, Monticello, Illinois, October 2023
30-minute research talk on host heterogeneity, social disparities, and impacts for SARS-CoV-2 control
Applied Math Graduate Seminar, University of Illinois at Urbana-Champaign, April 2022
25-minute research talk on social inequity and COVID-19 transmission MicroERA, University of Illinois at Urbana-Champaign, April 2022 Research speed talk on social inequity and COVID-19 transmission

POSTERS AT MEETINGS AND CONFERENCES

Ecology and Evolution of Infectious Diseases (EEID), Blacksburg, Virginia, June 2026
Same household, different choices: variation in health behaviors related to respiratory viruses in Illinois
Epidemics, San Diego, California, November 2025
Community connectivity and self-isolation as strategies to reduce the burden of respiratory diseases
American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, New Orleans, Louisiana, November 2024
Reimagining the serocatalytic model for seasonal coronaviruses
Institute for Disease Modeling (IDM) Annual Symposium, Seattle, Washington, September 2024
Reimagining the serocatalytic model for seasonal coronaviruses
Ecology and Evolution of Infectious Diseases (EEID), Stanford, California, June 2024
Disentangling the relative benefit of bivalent vs. monovalent boosters in the context of host immune history and socioeconomic status: a modeling study
MicroERA, University of Illinois at Urbana-Champaign, March 2024
Disentangling the seroconversion and seroreversion rates of seasonal coronaviruses using age-stratified seroprevalence data
Models of Infectious Disease Agent Study (MIDAS) Network Annual Meeting, Atlanta, Georgia, October 2023
Influence of immune history on booster impacts in low- and middle-income countries
ASTMH Annual Meeting, Chicago, Illinois, October 2023
Influence of immune history on booster impacts in low- and middle-income countries
MIDAS Network Annual Meeting, Bethesda, Maryland, September 2022
Funded by MIDAS to attend annual meeting and present on social inequity and COVID-19 transmission
Bohrer Memorial Student Workshop in Statistics, University of Illinois at Urbana-Champaign, April 2022
Poster presentation on social inequity and COVID-19 transmission

PUBLIC ENGAGEMENT

Science on Tap, Rigg's Brewery, November 2023
"Who gets sick and why: from COVID-19 to the common cold" - 1 hour discussion for a lay audience.
Genome Day, Champaign Public Library, November 2023
Two weekends of science exhibits and activities for a K-12 audience

SERVICE

Research Volunteer, The Ayah Project, October 2025 - present
Support research and education activities for the United States' first 501(c)(3) dedicated to supporting the LGBTQ+ Muslim community.
Campus Advisory Committee on the LGBTQ+ Communities at Cal, December 2025 - May 2026
Served as one of two postdoctoral appointees. This is a standing committee for the UC Berkeley senior administration, advising on concerns of lesbian, gay, bisexual, transgender, intersex, and queer UC Berkeley employees and students. Member of the Health and Wellness subcommittee, helping to develop policy recommendations in response to student and employee needs.
MIDAS Trainee Committee Co-Chair, November 2024 - January 2026
MIDAS Trainee Committee, January 2024 - January 2026
Organize events for trainees in the Models of Infectious Disease Agent Study (MIDAS) Network
Disability Caucus, Graduate Employees Organization, Spring 2021 - Spring 2023
With Disability Caucus, which advocates for disabled students and workers on campus, developed a workshop on disability and the accommodations process aimed at teaching assistants, staff, and faculty. The workshop ran in multiple departments including as part of math department TA training.
Accommodation Working Group, August 2022 - October 2022
Contributed to early stages of a working group formed by the Department of Mathematics to develop recommendations of best practices for accommodating disabled students in the graduate program

Undergraduate Affairs Committee, Fall 2021 - Spring 2022
Served on the committee as the graduate representative. Helped make decisions about undergraduate curriculum and programs.

AD-HOC REVIEWER

BMC Infectious Diseases; Nature Communications; Proceedings of the Royal Society B; Epidemics; Bulletin of the World Health Organization; Humanities and Social Sciences Communications; Journal of Health, Population, and Nutrition; Mathematical Biosciences; Scientific Reports

SELECTED COVERAGE

This Week in Virology, Episode 1035, August 2023: Poxvirus wars, warpspeed COVID-19 vaccine rollout
University of Illinois at Urbana-Champaign, School of MCB Communications, April 2024: Cheers to 'Science on Tap:' Graduate students demystify science topics at local brewery
Department of Statistics News, January 2023: Heat waves, stock prices, and COVID vaccines among student research projects presented at recent URES symposium